Key Xu

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Education

University of California, Santa Barbara — Santa Barbara, CA
B.S. Data Science | Minors: Computer Science — *Aug. 2023 – May 2025*

Virginia Polytechnic Institute and State University — Blacksburg, VA
B.S. Computer Science | Minors: Math, Statistics | Summa Cum Laude — *Aug. 2021 – May 2023*
GPA: 3.94 | 4.0 In-major, Rank 4, ICPC Regionals, VTHacks, RoboGrinder, COINS, Google Dev Club

Experience

Coinbase — July 2025 - Present
Software Engineer II — *New York City, NY*

- Refactored production market data processing library for trading application by eliminating per-tick heap allocs and lock contentions, reducing tick **processing latency by ∼25%** E2E in production (**C++**)
- Contributed benchmarking infrastructure for market data library, instrumenting order book tick processing path to establish latency baselines and validated **18-28% latency reduction** across optimizations (**C++**)
- **Top 1%** AI user. Built agent graph system for deploying custom autonomous long horizon workflows + org-wide memory layer for enforcing code quality and context savings. Used internally across various Coinbase. (**TS**)
- Readied equities launch in 3 weeks, fully refactored raise platform to support IPOs and other capital raising products, and built out Deribit migration flows (**TS, React, React Native, Go**)
- Built token sales platform, **0 to MVP** in 3 months, top code contributor, **$15 million** spot revenue in one week

Capital One — June. 2024 – Aug. 2024
Software Engineering Intern — *Dallas, TX*

- Migrated traditional loan payment batch processing system to AWS cloud through CDK, improving efficiency and **10x throughput**, and automating manual processing (**TypeScript, AWS Lambda, S3, Step Functions**)
- Integrated performance tests to monitor processing system latency and throughput (**JavaScript, K6 Grafana**)

SimonComputing — June 2023 – Aug. 2023
Software Engineering Intern — *Washington D.C.*

- Built a service layer for metadata migration bookkeeping for a data dictionary application, integrating Snowflake-backed lookups and audit/status tracking (**C++, Snowflake**)

Projects

Engram/Brain - Open Source Agentic Memory Layer MCP | *TypeScript, SQLite, Hugging Face* — May. 2026

- Solo developed lightweight, fully local memory MCP, droppable into anyone's agent for 80% context savings, and massive code quality gains, shareable knowledge, used internally at Coinbase (including CEO/CTO)

Consulting Slide Generator - Haize Labs (acquired) | *Python* — Feb. 2026

- Built a pixel precise powerpoint generator for largest bank in Singapore; able to categorize slide layouts, semantically allocate textual and visual data to create presentations, fully functional and shipped in 3 days

LegalPay - Founding SWE | *TypeScript, C++, Boost, PostGres, Docker, AWS* — Apr. 2025

- Architected **financial application** connecting LRS companies, clients, and attorneys, with realtime payments
- Built distributed backend with async processing; deployed MVP in **10 days** amid changing specs

NLP Calendar – Stanford University Treehacks | *Next.js, OpenAI, Postgre, Prisma, Supabase* — Feb. 2024

- Developed automatic calendar event generation using natural language, supporting groups with >**100** people, using Convex database and vector search (**TypeScript, OpenAI, Next.js, React**)

J.P. Morgan Data for Good - 1st Place | *Python* — Oct. 2023

- Engineered optimizer to output **top 10** cities for the MoveToProsper education program to target, to maximize program impact potential, improving program success rate by **30%** (**Python**)
- Applied PCA/MDS to **20+ datasets** (education, housing, income) to reduce dimensionality of superscore

GPU Speedup Analysis | *C, Bash, MPI, CUDA* — Mar. 2023

- Parallelized Lloyd's algorithm on MNIST dataset with MPI library, modeling ideal vs. actual speedup
- Implemented K-means approximation on NVIDIA GPUs through **CUDA**, achieving optimal ∼**16x** speedup

Technical Skills

Languages: C++, C, Java, JavaScript/TypeScript, Go, SQL, Assembly, R, Bash, HTML, CSS
Technologies: Node.js, AWS, PyTorch, Kubernetes, Docker, OpenMP, CUDA, Pandas, NumPy